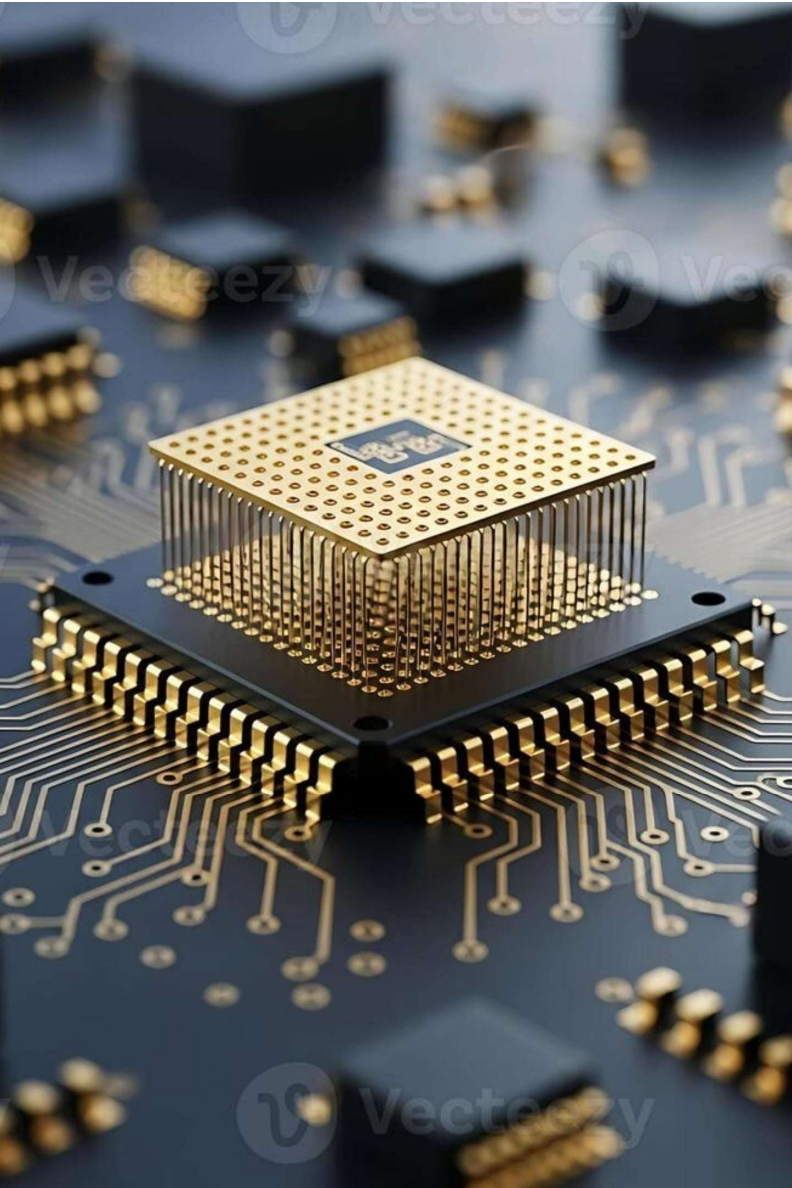




Evolution of Processor Architecture

From Intel 8085 to Intel Core & ARM

COMPUTER ARCHITECTURE – CSA1214
CO1 – AT3 – TECHNICAL PRESENTATION

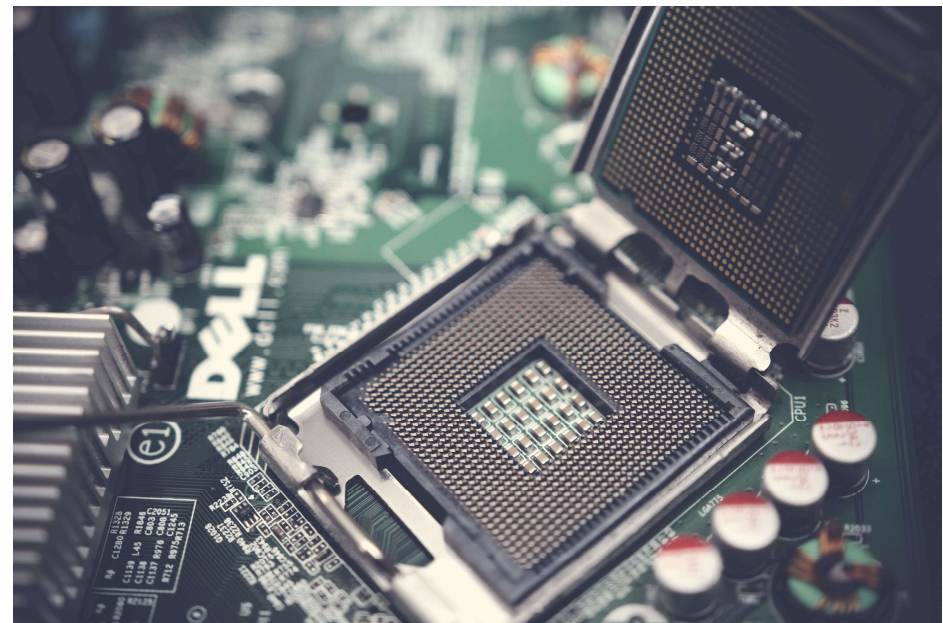
UNDER THE
GUIDANCE OF :
Dr. RADHIKA S

BY:
RAGAVI S
(192565027)
**B.E – CSE (CYBER
SECURITY)**

Abstract

Processor architecture defines how a CPU is designed and operates. This seminar traces the evolution from early 8-bit microprocessors to today's multi-core CPUs, covering the major milestones and future directions.

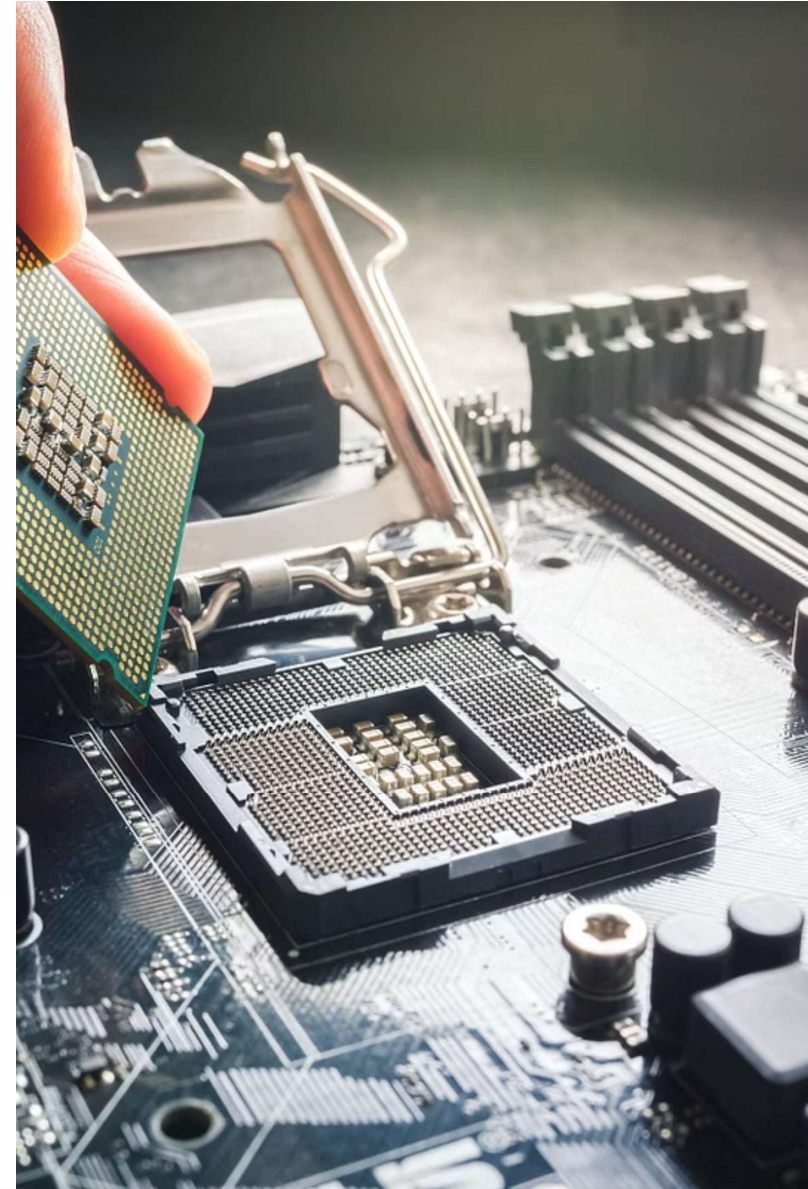
- Early 8-bit Era
- 16-bit & x86
- Multi-Core Age
- ARM & Efficiency
- Future Directions



Introduction: What Is Processor Architecture?

Processor architecture defines how a CPU is organized — its instruction set, data paths, memory hierarchy, and execution model. The CPU performs the **fetch–decode–execute cycle**, communicating with memory and I/O devices to run software.

- **ALU** — performs arithmetic and logic operations
- **Control Unit** — directs instruction execution flow
- **Registers & Cache** — ultra-fast on-chip storage



Intel 8085 — The 8-Bit Foundation (1976)

Architecture

8-bit data bus, 16-bit address bus, supporting up to **64 KB** of memory

Clock Speed

Operated at **3–5 MHz**, single +5V power supply

Key Features

Integrated serial I/O, five hardware interrupts, on-chip clock generator

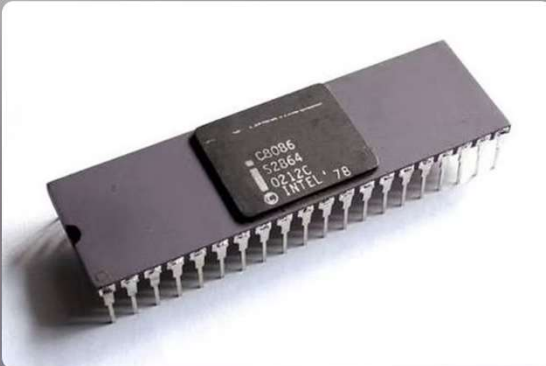
Applications

Embedded systems, industrial controllers, early personal computers, educational kits



The Intel 8085 became the cornerstone of early computer architecture education and embedded design worldwide.



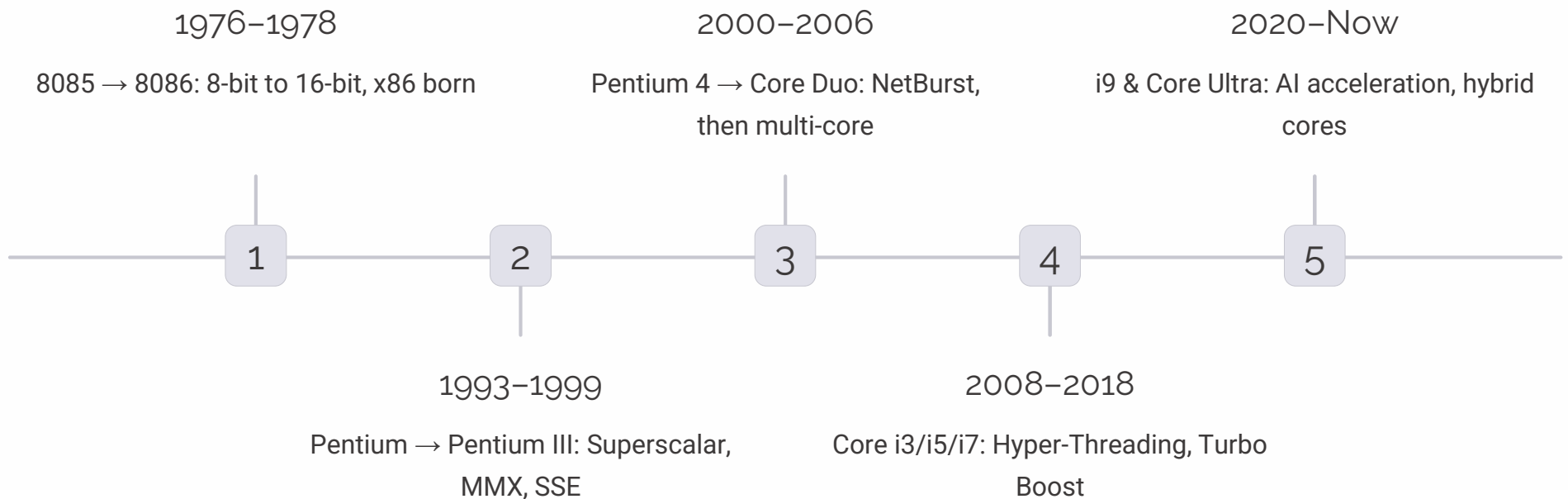


Intel 8086 — The 16-Bit Leap (1978)

The Intel 8086 introduced **16-bit architecture** and **segmented memory**, establishing the x86 foundation still used today.

- 20-bit address bus — Supporting up to **1 MB** of memory
- Segmented memory model — Code, data, stack, and extra segments
- Pipelined execution — Bus Interface Unit + Execution Unit
- x86 legacy — Basis for all modern Intel/AMD processors

Evolution of Intel Processors



Intel's biggest leaps have centered on **multi-core processing**, **Hyper-Threading**, **Turbo Boost**, and **AI acceleration**. Each generation has pushed higher performance while improving efficiency.

ARM Architecture — Efficiency by Design

ARM (Advanced RISC Machine) uses a Reduced Instruction Set (RISC) approach, prioritising low power consumption and energy efficiency over raw performance.



Ultra-Low Power

Optimised for battery-powered devices — smartphones, tablets, wearables



Embedded & IoT

Dominates embedded systems, microcontrollers, and IoT devices globally



Apple M-Series

Apple's transition to ARM-based M1/M2/M3 chips redefined laptop performance and efficiency



Data Centre

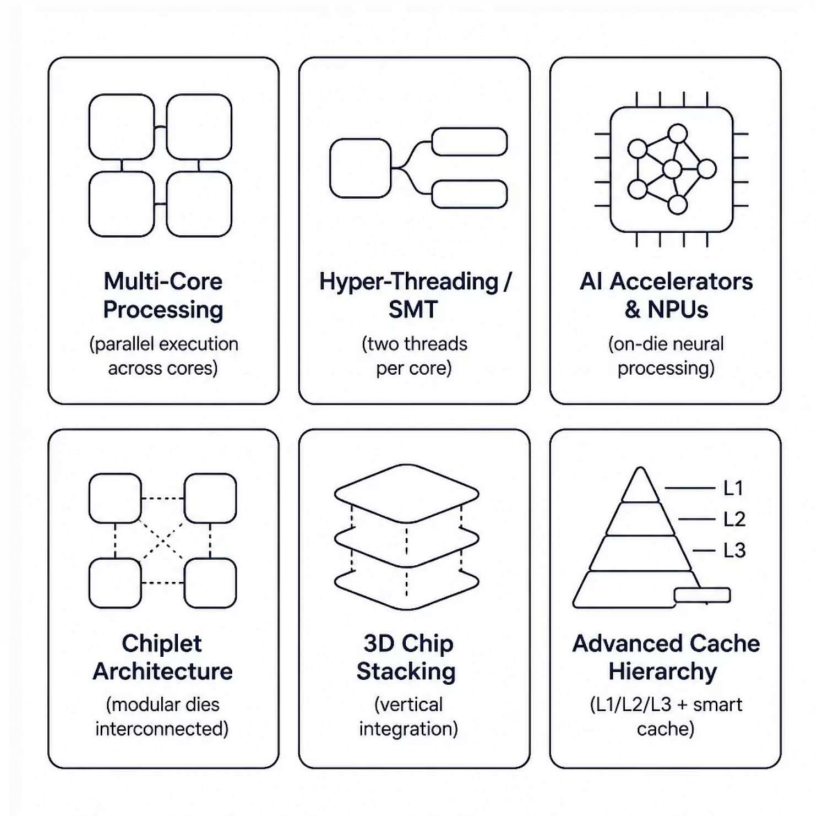
ARM servers gaining traction for cloud workloads requiring high efficiency



Intel Core vs ARM — Architecture Compared

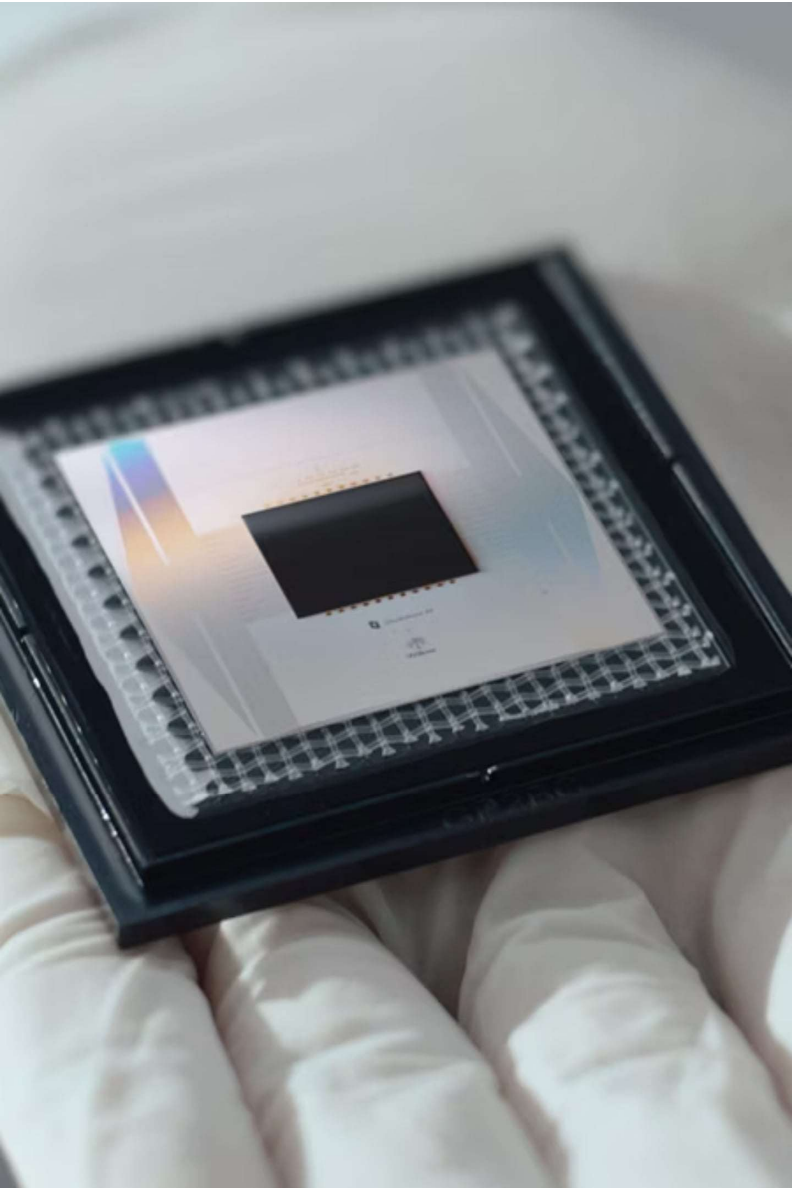
Feature	Intel Core (CISC)	ARM (RISC)
Architecture	Complex Instruction Set	Reduced Instruction Set
Performance	High — desktops & servers	Efficient — mobile & embedded
Power Consumption	Higher (15–125W)	Very low (1–10W)
Battery Life	Moderate	Excellent
Primary Use	PCs, workstations, servers	Smartphones, tablets, IoT
Notable Example	Intel Core i9, Core Ultra	Apple M3, Snapdragon

Modern Processor Technologies



Today's processors combine more cores, smarter threading, and dedicated AI hardware to boost throughput without wasting power.

They also use chiplet designs, 3D stacking, advanced cache, and dynamic power management to improve efficiency, scalability, and responsiveness.



The Future of Processor Architecture



Neuromorphic Chips

Brain-inspired processors mimicking neural networks for ultra-efficient AI



Quantum Computing

Qubit-based processors solving problems beyond classical CPU capability



Photonic Processors

Light-based data transmission for near-zero latency and heat



Green Computing

Ultra-low-power architectures and sustainable chip design for a carbon-neutral future



Key Takeaway: From the 8-bit Intel 8085 to AI-accelerated Core Ultra and ARM-based M-series — processor architecture continues to evolve, driven by demands for performance, efficiency, and intelligence. **Thank You.**

THANK YOU!!!